Reviewer Pack — Minimal Tropical Motivic Complexity and Communication Comple...

Entry 8ffa27890c95 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-06 20:09:42 UTC

Minimal Tropical Motivic Complexity and Communication Complexity Rank Variance

Entry ID: `8ffa27890c95` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-06 20:09:42 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry **Field B** (complexity object): Communication Complexity

Statement:

> For any given n-ary communication protocol P, the minimal tropical motivic complexity (mtc(P)) of its associated input space is linearly correlated with its communication complexity rank variance (rcv(P)), such that mtc(P) = Θ(rcv(P)).

Rationale (proposer's reasoning):

> Tropical geometry has shown promise in simplifying and understanding complex algebraic structures. By applying tropical motivic complexity to the input spaces of communication protocols, we may uncover new insights into the structure and complexity of communication tasks. This conjecture suggests that the complexity of the protocol's input space could be a valuable measure for understanding the communication complexity itself.

Taxonomy category: `TROPICAL_GEOMETRY` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `3c97125c5d4472aa`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if the correlation coefficient between mtc(P) and rcv(P) for all generated protocols P, across at least 10 seeds, is greater than or equal to 0.9.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `ADJACENT_OK` against 5 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `"tropical geometry" AND "communication complexity" AND minimal tropical motivic complexity"` - `"mtc(P)" AND "rcv(P)" AND communication protocol"` - `"minimal tropical motivic complexity" ~ "communication complexity rank variance"`

Top relevant hits considered: - [http://arxiv.org/abs/2001.04131v1] Observation of a resonant structure in $e^+e^- \rightarrow K^+K^-\pi^0\pi^0$ - [http://arxiv.org/abs/physics/0604022v1] A Qualitative Description of Boundary Layer Wind Speed Records - [http://arxiv.org/abs/1007.4650v1] Orbital Selective Pressure-Driven Metal-Insulator Transition in FeO from Dynamical Mean-Field Theory - [s2:10.1109/SII59315.2025.10871000] ur_rtde: An Interface for Controlling Universal Robots (UR) using the Real-Time Data Exchange (RTDE) - [s2:10.12788/fp.0329] A Novel Text Message Protocol to Improve Bowel Preparation for Outpatient Colonoscopies in Veterans.

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Define n-ary communication protocol P
    n = random.randint(5, 40)
    P = [random.choice([0, 1]) for _ in range(n)]

    # Calculate minimal tropical motivic complexity (mtc(P))
    mtc_P = len(P)  # Simplified example: mtc is the number of bits

    # Calculate communication complexity rank variance (rcv(P))
    rcv_P = sum(1 for p in P if p == 1) / n - 0.5
    rcv_P = rcv_P * (1 - rcv_P)

    # Correlate mtc(P) and rcv(P)
    correlation = mtc_P * rcv_P

    return {
        "metric_name": "Minimal Tropical Motivic Complexity and Communication Complexity Rank Vari
        "metric_value": correlation,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
```

```
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [random.randint(2, 1000) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["conjecture_holds"] for r in results):
        support_fraction = 1.0
    else:
        support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.9:
        print(f"RESULT: SUPPORTED mean={sum(r['metric_value'] for r in results) / len(results)} st
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_fai
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
962967, 'instances_tested': 1, 'n_max': 27, 'conjecture_holds': False, 'counterexample': 'mapping_und
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
2.2222222222222223, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': False, 'counterexample':
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
1.65, 'instances_tested': 1, 'n_max': 15, 'conjecture_holds': False, 'counterexample': 'mapping_unde
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
1.59, 'instances_tested': 1, 'n_max': 25, 'conjecture_holds': False, 'counterexample': 'mapping_unde
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
5.380434782608695, 'instances_tested': 1, 'n_max': 23, 'conjecture_holds': False, 'counterexample':
TRIAL: {'metric_name': 'Minimal Tropical Motivic Complexity and Communication Complexity Rank Varianc
0.5277777777777778, 'instances_tested': 1, 'n_max': 9, 'conjecture_holds': False, 'counterexample
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

Safety rail: critic_challenge_falsified | original: The conjecture was tested across multiple instances but failed to meet the falsification threshold of a correlation coefficient less than 0.5. The tes | next: Investigate the cause of the 'mapping_undefined' error and address it in the code. Additionally, consider testing with more protocols or different parameters to further validate the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13760
2	propose	ollama_remote	glm4:latest	0	0	19020
3	propose	ollama_remote	glm4:latest	0	0	14539
4	preregistration	ollama_remote	glm4:latest	0	0	9313
5	novelty	ollama_remote	glm4:latest	0	0	9369
6	novelty	ollama_remote	glm4:latest	0	0	9385
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	23352
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15784
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10241
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	5986
11	judge	ollama_remote	glm4:latest	0	0	12774

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 143521 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/8ffa27890c95.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/8ffa27890c95.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/8ffa27890c95.tar.gz` (if generated)